SHIVAM KAMBOJ — RESEARCH SUMMARY

Ph.D. Physics · expected Fall 2026

Light–matter Hamiltonians · open-system simulation · qudit control and tomography · surface-code simulation

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HOW THE PARTS OF MY WORK CONNECT

Model the Hamiltonian → Simulate it → Control and characterize → Error-correct

1. HAMILTONIAN MODELING
- **Light–Matter Hamiltonians**
- Normal-Mode & Bogoliubov
- Parity & Exchange Symmetry
- True vs. Avoided Crossings
- **Rabi Dimer · YIG–Cavity**

2. NUMERICAL SIMULATION
- **Sparse Exact Diagonalization**
- Lindblad Master Equations
- Fock-Space Convergence
- COMSOL RF/AC-DC Extraction
- **QuTiP · SciPy · SLURM**

3. CONTROL & CHARACTERIZATION
- **Pulse-Level Qudit Control**
- Linear-Inversion QPT Pipeline
- Pseudo-Inverse Reconstruction
- Automated Ramsey & T_1/T_2 Fit
- **LLNL Internship, 2024**

4. ERROR CORRECTION
- **Correlated $1/f$ & OU Noise**
- Stim Circuit Sampling
- PyMatching MWPM Decoding
- Bootstrap Threshold Estimates
- `nonmarkov_qec` **(PyPI)**

SELECTED WORK

Colors match the stages above

STAGES 1–2 · MODELING & SIMULATION

Quantum Rabi Dimer (Tian Group, UC Merced)

Model ultrastrong light–matter coupling in the quantum Rabi dimer. Normal-mode analysis and sparse exact diagonalization, resolved by parity and site-exchange symmetry, separate true from avoided level crossings and identify a candidate cat-state doublet. Lindblad simulations in QuTiP run as checkpointed SLURM jobs with convergence checks in Hilbert-space truncation. COMSOL models of a proposed superconducting implementation give capacitances, impedances, and inductive participation. I believe the doublet's parity structure can be extended to a noise-biased bosonic qubit, which I plan to test next.

STAGE 3 · CONTROL & CHARACTERIZATION

Qudit Control and Process Tomography (LLNL, 2024)

Wrote Python software for pulse-level control of multilevel superconducting qudits (transitions up to $|3\rangle$) and evaluated optimal-control gate sequences. Built a generalized linear-inversion process-tomography pipeline, using Moore–Penrose pseudo-inverse reconstruction, that turns measured data into process maps. Automated Ramsey and T_1/T_2 fits to give quantitative feedback during control experiments.

STAGE 4 · ERROR CORRECTION

Surface Codes under Correlated Noise (`nonmarkov_qec`)

Open-source Python package, released on PyPI. Generates Ornstein–Uhlenbeck and sum-of-OU ($1/f$) dephasing, injects it into precompiled Stim circuits (about $35\times$ faster than rebuilding them), decodes with PyMatching, and estimates thresholds with parametric-bootstrap 95% confidence intervals on rotated surface codes ($d = 3, 5$). In the regime studied, correlated noise did not lower the threshold relative to a matched white-noise baseline, consistent with Kam et al. (2025). 74 tests, strict type checking, and CI.

STAGE 1 · OTHER MODELING & SOFTWARE

Magnon–Photon Systems, Spin-Torque Arrays, PCAT

Develop Hamiltonian models of coupled YIG–cavity systems with parametric driving, working with an experimental group (Sharping Lab). Earlier, derived dispersion relations and numerical solvers for 2D spin-torque oscillator arrays and studied edge modes under disorder (*Phys. Rev. B* **109**, 094436, 2024). Also built PCAT, a physics-teaching web app on Google App Engine (Python, Flask, Gemini API), piloted in UC Merced courses.

HOW THIS RELATES TO DIFFERENT PLATFORMS

Direct = done on that platform · Related = shared physics and methods

Platform (examples)	Shared physics	My relevant work
SUPERCONDUCTING **Google, IBM, Rigetti, SLAC**	Circuit-QED Hamiltonians, anharmonic qudit levels, $1/f$ flux & charge dephasing.	**Direct:** qudit pulse control and linear-inversion tomography software (LLNL); COMSOL modeling of a proposed circuit; surface-code simulation under $1/f$ noise.
NEUTRAL ATOM & ION **Atom, Oratomic, Quantinuum**	Rabi-type light–matter coupling; motional and Rydberg modes; laser phase and intensity noise.	**Related:** Rabi-model Hamiltonians; exact-diagonalization and Lindblad simulation; surface-code simulation under correlated dephasing.
PHOTONIC & SILICON SPIN **PsiQuantum, Quantum Motion**	Bosonic Fock-space truncation and photon loss; $1/f$ charge noise in spin devices.	**Related:** parity-resolved modeling of bosonic cat states with photon loss; $1/f$ noise models; Stim/PyMatching Monte Carlo simulation.
CONTROL & SEMICONDUCTOR **Quantum Machines, SDT, AMAT**	Pulse-level control; microwave impedances; electromagnetic simulation.	**Direct:** pulse-level control, tomography, and Ramsey/T_1/T_2 fitting software (LLNL); finite-element RF modeling in COMSOL.